UNIVERSITI TEKNOLOGI MARA
FINAL EXAMINATION

COURSE	PROGRAMMING I
COURSE CODE	CSC402
EXAMINATION	NOVEMBER 2026
TIME	3 HOURS

INSTRUCTIONS TO CANDIDATES

1. This question paper consists of the questions printed in this booklet.
2. Answer ALL questions in the Answer Booklet. Start each answer on a new page.
3. Do not bring any material into the examination room unless permission is given by the invigilator.
4. Please check to make sure that this examination pack consists of:
   i) the Question Paper
   ii) an Answer Booklet - provided by the Faculty
5. Answer ALL questions in English.

DO NOT TURN THIS PAGE UNTIL YOU ARE TOLD TO DO SO

This examination paper consists of 5 printed pages.

PART A (10 MARKS)

Answer ALL questions.

1. Which phase of the program development life cycle defines what the program must do?
   A. Analysis
   B. Testing
   C. Coding
   D. Maintenance

2. Which part of a for loop is executed after each iteration?
   A. The condition only
   B. The update expression
   C. The loop body twice
   D. The initialisation

3. What does the % operator return?
   A. The quotient
   B. The square root
   C. The remainder of integer division
   D. The absolute value

4. Which operator sends data to cout?
   A. <<
   B. ==
   C. >>
   D. =

5. Which data type stores a whole number?
   A. float
   B. string
   C. int
   D. char

6. Which structure chooses between two alternative paths?
   A. for
   B. struct
   C. if...else
   D. array

7. Which identifier is valid in C++?
   A. total_score
   B. total score
   C. 2total
   D. int

8. Which declaration creates an integer array of five elements?
   A. int a[5];
   B. int a(5);
   C. int a;
   D. array a[5];

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9. Which item is NOT a programming paradigm?
   A. Debugging
   B. Procedural programming
   C. Object-oriented programming
   D. Structured programming

10. Which statement immediately exits a loop?
   A. break
   B. cin
   C. return 0
   D. continue

PART B (60 MARKS)

QUESTION 1

This question tests programming concepts and error terminology.

a) Define compiler and interpreter.

(2 marks)

b) State THREE differences between a compiler and an interpreter.

(3 marks)

QUESTION 2

A program reads three integer values and displays the largest value.

a) Draw a flowchart to find the largest of three values.

(5 marks)

b) Write pseudocode and a complete C++ program for the flowchart.

(8 marks)

QUESTION 3

This question tests declarations, constants and expressions.

a) Write C++ declarations for: an integer age, a double price, a constant discount rate of 0.10 and a character grade.

(6 marks)

b) Write C++ statements to read age and price, calculate price after discount, and display the result.

(5 marks)

QUESTION 4

Study the following C++ program segment:

```
int total = 0;
for (int i = 1; i <= 5; i++) {
   total += i;
}
cout << total;
```

a) Trace the program and state the output.

(10 marks)

b) Rewrite the loop using a while loop. Then explain what changes if the condition is i < 5.

(5 marks)

QUESTION 5

A sports club stores sport names and medal counts in parallel arrays.

a) Declare two parallel arrays for five sports and initialise them with suitable values.

(10 marks)

b) Write a program segment to display each sport with its medals, then calculate the total and average medals.

(6 marks)

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PART C (30 MARKS)

QUESTION 1

A lecturer records the marks of ten students in an array.

Write a complete C++ program that reads ten marks, displays the average, the highest mark and its index, and counts how many marks are above the average.

(15 marks)

QUESTION 2

A bookstore records up to five books. Each book has a title, price and quantity in stock.

Define a struct Book. Write a complete C++ program that reads the records, calculates the total inventory value, and displays the title of the most expensive book.

(15 marks)

END OF QUESTION PAPER